Alvaro Valdez-Duran

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Education

University of Washington **June 2025**
Bachelor of Science in Computer Science & Systems *Tacoma, Washington*

Relevant Coursework

- Computer Security
- Security Engineering
- File Scripting
- Data Structures
- Database Management
- Web Development
- Object-Oriented Programming
- Machine Learning
- Operating Systems

Experience

University of Washington **September 2024 - March 2025**
Teaching Assistant *Tacoma, Washington*

- Assisted over 30 students in understanding core programming concepts like variables, loops, and conditionals in an introductory programming course, leading to a 20% increase in average student performance on assignments.
- Provided hands-on support during weekly lab sessions, troubleshooting and guiding students through debugging processes, improving their skills in solving problems and code quality.
- Worked with provided lab materials to facilitate weekly lab sessions, increasing student engagement and improving the retention of key concepts by 15%.
- Utilized strong communicative skills in conveying complex concepts to students in an understandable manner to resolve issues.

Projects

Library Book Management System | *React, Node.js, Next.js, PostgreSQL, Docker, Postman, Python* **Oct - Dec 2024**

- Collaborated in a team of four to develop a secure full-stack library management system, applying Agile practices (sprint planning, code reviews) to enhance delivery speed and maintain code quality.
- Implemented backend API routes in Node.js with role-based access control (RBAC) and input validation to prevent unauthorized access and injection attacks, ensuring data integrity across PostgreSQL.
- Configured secure Dockerized environments to ensure consistent deployments and minimize misconfiguration risks, aligning with DevSecOps best practices.
- Integrated logging and monitoring for API endpoints using Postman and custom scripts to simulate alerting workflows, reflecting SOC-style visibility into application activity.

Dungeon Adventure | *LibGDX, Java* **July - August 2024**

- While developing a Java-based dungeon exploration game in LibGDX with a team of three, I implemented core gameplay mechanics, including player movement and combat systems. By ensuring seamless integration of game logic, we improved responsiveness and reduced movement lag by 40%, enhancing the overall player experience.
- Contributed to 30% of core systems, ensuring seamless integration between player actions, combat logic, and visual/audio feedback.
- To ensure timely delivery of project milestones in a fast-paced environment, I coordinated with teammates using Agile methodologies like daily stand-ups and sprint planning. This improved task prioritization and reduced project delays by 20%, ensuring all deadlines were met efficiently.

Stuxnet Cybersecurity IR Analysis | *Malware Analysis, Threat Intelligence, MITRE ATT&CK* **March 2025**

- Performed a comprehensive incident response analysis of the Stuxnet malware, mapping each phase of the attack lifecycle to the MITRE ATT&CK framework and identifying key TTPs used by the threat actors.
- Researched Advanced Persistent Threat (APT) tactics and documented exploitation of zero-day vulnerabilities (CVE-2010-2568, CVE-2010-2729, CVE-2008-4250, CVE-2010-2743) enabling infiltration of air-gapped industrial systems.
- Assessed defensive measures and response strategies, evaluating Iran's cyber resilience improvements and lessons learned for modern ICS/SCADA environments.
- Produced an executive-level incident report and delivered a technical briefing detailing attack vectors, detection opportunities, and mitigation recommendations.

Technical Skills

Security Tools: Wazuh, Wireshark, Postman, Docker, Snort
Languages: Python, Java, JavaScript, TypeScript, C, C#, SQL, R
Frameworks / Technologies: React, Next.js, Node.js, LibGDX, Arduino
Developer Tools: VS Code, IntelliJ, PyCharm, Git, GitHub
Cloud Platforms: AWS, Azure, Render